Can Machine Learning Beat the Historical Average? Forecasting the US Equity Premium with Regularized Models, Technical Indicators and Switching Strategies

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Abstract

We ask whether regularized machine learning models, fed with the fourteen Goyal–Welch macroeconomic predictors (one to six lags) and the fourteen technical indicators of Neely et al. (2014), can forecast the monthly simple US equity premium better than the prevailing historical mean. Using a fixed 240-month rolling window on 1951:02–2025:12 data, we evaluate seven models and two forecast combinations at horizons of 1, 3 and 12 months. Only ridge regression attains a positive out-of-sample R^2 (0.12% at $h = 1$ and 0.30% at $h = 3$), and none of the 54 one-sided Diebold–Mariano tests rejects equal accuracy (smallest $p = 0.23$). An adapted replication of Neely et al. (2014) yields positive but insignificant single-signal gains on 1966–2011 and uniformly negative ones on 2012–2025. Economically, a ridge-driven switching strategy ends at \$499 per dollar versus \$342 for buy-and-hold at $h = 1$ net of 10 bps costs, but four of nine forecast strategies trail buy-and-hold, most strategies lose at $h = 12$, and these gains are not tested for significance. Overall, we find no statistically reliable improvement on the historical average.

1 Introduction

Whether the equity premium is predictable out-of-sample matters both for asset allocation and for asset pricing theory. Goyal and Welch (2008) showed that a broad set of macroeconomic predictors fails to beat the prevailing (historical) mean in real time, and Goyal, Welch and Zafirov (2024) reach a similar verdict with a much larger predictor set. Campbell and Thompson (2008) argue that even a small positive out-of-sample R^2 can be economically valuable, while Neely, Rapach, Tu and Zhou (2014) report that technical indicators contain out-of-sample information on their 1966–2011 sample. Machine learning appears to offer a way forward: Gu, Kelly and Xiu (2020) document gains from shrinkage and nonlinear methods in asset pricing. Monthly premium forecasting, however, combines a very low signal-to-noise ratio with many correlated, persistent predictors—exactly the setting in which flexible models can overfit.

We address three research questions. *RQ1*: can regularized and dimension-reduction models (or forecast combinations) produce a positive and statistically significant out-of-sample R^2 against the prevailing mean? *RQ2*: do the Neely et al. (2014) technical signals add predictive value, on their original sample and after 2011? *RQ3*: do the forecasts deliver economic value to a switching investor or a mean–variance investor after transaction costs?

Our contribution is a carefully time-aligned comparison rather than a new model. We rebuild the fourteen technical signals from raw price and volume data, combine them with lagged macro

predictors in a 98-column design, estimate seven models in a fixed rolling window with horizon-correct in-window tuning, use the prevailing mean as the only benchmark, and report both statistical and economic evaluations, including several robustness checks. Several design choices were revised after the instructor's feedback on our proposal (Section 3.5).

The answers are largely negative. Only ridge regression has a positive R^2_{OOS} (0.12% at $h = 1$, 0.30% at $h = 3$), no Diebold–Mariano test is significant at any horizon, and the random forest is the least accurate model throughout. The single-signal technical regressions are insignificant on 1966–2011 and negative afterwards. Some switching strategies end with more wealth than buy-and-hold at $h = 1$, but the gains are model-specific, shrink at longer horizons, and are not supported by a significance test.

2 Data

2.1 Target and predictors

The target is the monthly *simple* equity premium in percent, $y_t = 100\,(R^m_t - R^f_t)$, where R^m_t is the S&P 500 return including dividends and R^f_t the risk-free rate, both taken from the continuously updated Goyal–Welch–Zafirov (2024) workbook. Over 1951:02–2025:12 the premium averages 0.66% per month with a standard deviation of 4.21% (Table 1). The fourteen macroeconomic predictors follow Goyal and Welch (2008): the dividend–price ratio, dividend yield, earnings–price ratio, dividend payout ratio, stock variance, book-to-market, net equity expansion, Treasury bill rate, long-term yield, long-term return, term spread, default yield spread, default return spread and inflation. Each enters with lags one to six, giving 84 columns.

The fourteen technical indicators follow Neely et al. (2014): six moving-average rules MA(s, l) with $s \in \{1, 2, 3\}$ and $l \in \{9, 12\}$, two momentum rules MOM(m), $m \in \{9, 12\}$, and six on-balance-volume rules VOL(s, l). Each is a binary buy signal. MA and MOM rules use the index level from the Goyal–Welch–Zafirov workbook; VOL rules use monthly average daily S&P 500 volume, taken from the Neely et al. replication files for 1950–2011 and from Yahoo Finance for 2012–2025. Against the published signals over 1950:12–2011:12 (733 months per signal), twelve signals match exactly and MOM(12) and VOL(2,9) differ in one month each (1968:03 and 1980:04), an overall match of 99.98%; we do not know the cause of the two differences. The baseline design therefore has 98 predictors (84 macro-lag columns and 14 signals).

2.2 Expanded predictor set

As a robustness variant we add eight variables from the Goyal et al. (2024) set (skvw, tail, rdsp, avgcor, dtoy, dtoat, wtexas, lzrt), each with six lags, giving 146 predictors. These are the only additional variables that are observed monthly throughout the sample and are not flagged as full-sample constructions. We exclude fbm, ogap, sntm, tchi and shtint because the data documentation describes them as full-sample constructions, so lagging them would not remove look-ahead bias; twelve further variables (e.g. cay, pce, eqis) are not available monthly; and csp, vp, impvar, vrp, ndrbl, ygap, rsvix and disag start after or end before our sample, so including them would shorten the common sample. The expanded variant is therefore a partial, not exhaustive, use of the Goyal et al. (2024) predictor set.

2.3 Alignment and sample

Row t of the panel holds the premium earned in month t and predictors observable at the end of month t: macro lag columns satisfy $x_t^{(k)} = x_{t-k}$, and technical signals use prices and volume through month t only. The common sample runs from 1951:02 (all lags and signals available) to 2025:12, giving $T = 899$ months. Table 1 reports summary statistics and augmented Dickey–Fuller tests: the premium is clearly stationary, whereas valuation ratios and interest-rate levels are highly persistent, which is one reason we rely on regularization and dimension reduction rather than unrestricted regression.

Table 1: Summary statistics and unit-root tests (sample: 1951:02–2025:12).

Series	Mean	SD	Min	Max	Skew	Kurt	ADF	Sig.
Panel A: simple equity premium (%, monthly)								
y_t (full sample)	0.662	4.211	-22.18	16.30	-0.42	4.54	-29.18	1%
Panel B: macroeconomic predictors (raw series)								
d/p	0.030	0.013	0.01	0.07	0.69	2.91	-2.82	>10%
d/y	0.030	0.013	0.01	0.07	0.69	2.91	-2.70	>10%
e/p	0.063	0.026	0.01	0.15	0.98	3.58	-3.57	5%
d/e	0.500	0.283	0.29	3.97	8.48	88.54	-5.92	1%
svar	0.002	0.005	0.00	0.07	10.87	149.96	-19.89	1%
b/m	0.486	0.250	0.12	1.21	0.70	2.72	-1.99	>10%
ntis	0.011	0.020	-0.06	0.05	-0.62	2.88	-4.19	1%
tbl	0.041	0.030	0.00	0.16	0.92	4.25	-2.87	>10%
lty	0.057	0.028	0.01	0.15	0.83	3.30	-1.83	>10%
ltr	0.005	0.027	-0.11	0.15	0.52	6.05	-28.18	1%
tms	0.016	0.014	-0.04	0.05	-0.04	2.75	-4.74	1%
dfy	0.009	0.004	0.00	0.03	1.89	8.25	-4.39	1%
dfr	0.000	0.014	-0.10	0.07	-0.66	10.97	-23.79	1%
infl	0.003	0.004	-0.02	0.02	0.14	5.54	-3.19	10%
Panel C: technical signals (binary; mean = fraction of months in "buy")								
MA_1_9	0.709							
MA_1_12	0.729							
MA_2_9	0.710							
MA_2_12	0.730							
MA_3_9	0.715							
MA_3_12	0.731							
MOM_9	0.729							
MOM_12	0.743							
VOL_1_9	0.694							
VOL_1_12	0.720							
VOL_2_9	0.688							
VOL_2_12	0.719							
VOL_3_9	0.704							
VOL_3_12	0.718							

ADF: intercept and trend, common-sample BIC lag selection; MacKinnon (2010) asymptotic critical values $-3.959/-3.410/-3.127$ (1%/5%/10%). $T = 899$ monthly observations before ADF lags.
Skewness and kurtosis are population moments; kurtosis is NOT excess (a normal distribution has kurtosis 3).
The premium is $100\,(R^m - R^f)$ using monthly simple returns; macro predictors enter the models with one to six lags; technical signals are the 14 binary rules of Neely et al. (2014).

3 Methodology

3.1 Forecasting design

At the end of month t we forecast the premium earned in month $t + h$ for $h \in \{1, 3, 12\}$. The target is always a single month's premium h months ahead, not a cumulative h-month return. A training pair links predictors X_s with the outcome y_{s+h}, and at origin t only pairs with $s+h \leq t$ are eligible, so no outcome after the origin is used. Every model is re-estimated each month on a fixed window of the 240 most recent eligible pairs (20 years). The out-of-sample periods contain 658 targets from 1971:03 at $h = 1$, 654 from 1971:07 at $h = 3$, and 636 from 1973:01 at $h = 12$. The benchmark is the prevailing mean, the expanding average of all premia observed up to the origin.

3.2 Models and tuning

The design is wide relative to a 240-month window and its columns are highly correlated, so we use methods covered in the course that regularize or compress the predictor space, plus one nonlinear learner. The seven models are: (i) LASSO with the data-driven plug-in penalty of `hdm::rlasso` (Chernozhukov, Hansen and Spindler, 2016), which sets the penalty from theory and heteroskedasticity-robust loadings instead of cross-validation; (ii) post-LASSO, an OLS refit on the rlasso-selected variables; (iii) ridge and (iv) elastic net ($\alpha = 0.5$) from `glmnet`; (v) principal components regression and (vi) partial least squares on standardized predictors with at most 20 components (zero components, i.e. the window mean, is also allowed); and (vii) a random forest with 500 trees, $m_{\text{try}} = \lfloor p/3 \rfloor$, minimum node size 5 and a fixed seed.

The penalty of ridge and elastic net and the number of components of PCR and PLS are chosen inside each training window by forward chronological validation. The second half of the window is split into three consecutive validation blocks; for each block the model is fitted only on earlier pairs whose outcomes would have been observed when the block starts (the $h - 1$ pairs immediately before the block are dropped), and squared validation errors are summed over the three blocks. Preprocessing (dropping constant columns, standardization) is redone within each fold, and the glmnet penalty grid is set from the first training segment only. The selected tuning parameter is then refitted on the full window. No information from the test month or later enters any forecast.

3.3 Forecast combinations

We combine the seven model forecasts with (i) an equal-weight average and (ii) a constrained Granger–Ramanathan (1984) regression. For the latter, at each origin the weights solve $\min_w \|y - Fw\|^2$ subject to $\sum_i w_i = 1$ and $w_i \geq 0$, where F contains only past out-of-sample forecasts whose errors are known at that origin. The constraints turn the regression into a convex weighting that cannot take offsetting long–short positions in highly correlated forecasts. Until 60 such observations exist, equal weights are used.

3.4 Evaluation

Accuracy is measured by RMSE, MAE, hit rate and the Campbell–Thompson (2008) statistic $R^2_{\text{OOS}} = 1 - \sum e_t^2 / \sum e_{t,\text{PM}}^2$ relative to the prevailing mean. Significance is assessed with one-sided Diebold–Mariano (1995) tests on squared and absolute error differentials (alternative: the model is more accurate), using a Newey–West variance with lag $h-1$ and normal critical values. Following the course, we use the Diebold–Mariano test only and do not report the Clark–West (2007) nested-model test.

Economic value is assessed in two ways. First, a binary switching strategy holds the S&P 500 in a month if the forecast of that month's premium is positive and the risk-free asset otherwise; we report terminal wealth per \$1, the annualized Sharpe ratio, maximum drawdown and the number of switches, and compare with buy-and-hold. In the main specification the position for month t uses the forecast of month t made at $t-h$, so for $h > 1$ the signal is older; Section 4.4 reports an alternative timing. Every strategy starts in cash and is charged the cost on every switch, including the initial equity purchase, with no terminal liquidation. Costs are 10 bps per switch in the baseline, with 5 and 20 bps as sensitivity cases. We assume the investor trades an S&P 500 ETF and Treasury bills; major US brokers now charge zero online commission on both (e.g. Charles Schwab, 2026), and a one-cent quoted spread on a liquid S&P 500 ETF priced in the hundreds of dollars is well below 1 bp. The 10 bps baseline is therefore a conservative all-in cost covering spread, slippage and price impact, and the 20 bps case reflects the higher trading costs of earlier decades. Second, a mean–variance investor with relative risk aversion $\gamma = 3$ holds $w_t = \hat{y}_t/(\gamma\hat{\sigma}_t^2)$ in equities, bounded to $[0, 1.5]$, where $\hat{\sigma}_t^2$ is the sample variance of the 60 monthly returns up to the origin. We report the annualized certainty-equivalent gain ΔU over the same investor using the prevailing-mean forecast, net of the same proportional costs on weight changes.

3.5 Changes after proposal feedback

Following the instructor's comments on our proposal, we replaced an OLS benchmark with the prevailing mean, replaced a glmnet LASSO with rlasso and post-LASSO, replaced the Clark–West test with Diebold–Mariano tests on squared and absolute errors, extended the macro predictors from one lag to up to six lags, and added the switching strategy and the constrained Granger–Ramanathan combination.

4 Results

4.1 Statistical accuracy and significance

Table 2 and Figure 1 summarize point-forecast accuracy. Of the 27 model–horizon pairs, only two have a positive R^2_{OOS}: ridge at $h = 1$ (0.12%) and at $h = 3$ (0.30%). At $h = 12$ every model is less accurate than the prevailing mean; the equal-weight combination comes closest (-0.48%). The random forest is the least accurate model at every horizon (-15.53%, -12.33% and -10.47%), consistent with a flexible learner fitting noise in a 240-month window. The simple combinations are generally close to the benchmark: the Granger–Ramanathan combination has

R^2_{OOS} of -0.54% at $h = 1$ and the equal-weight combination -0.14% at $h = 3$.

None of these differences is statistically significant. Across nine models, three horizons and two loss functions (54 tests), no one-sided Diebold–Mariano test rejects equal accuracy even at 10%. The smallest p-value is 0.23, for ridge at $h = 1$ under absolute-error loss; under squared-error loss ridge has $p = 0.451$ at $h = 1$ and $p = 0.341$ at $h = 3$. The positive ridge R^2 values should therefore be read as indistinguishable from zero, particularly as they are the best of 27 comparisons. Hit rates lie between 54.1% and 61.2%, against 59.7% to 59.9% for the prevailing mean, so directional accuracy is also close to the benchmark.

Figure 2 shows that the Granger–Ramanathan combination at $h = 1$ concentrates on a few linear models—mainly ridge, PCR and elastic net, with ridge dominant after 2009—and gives essentially no weight to the random forest. This reflects which models had smaller past errors, but it did not translate into a significant gain.

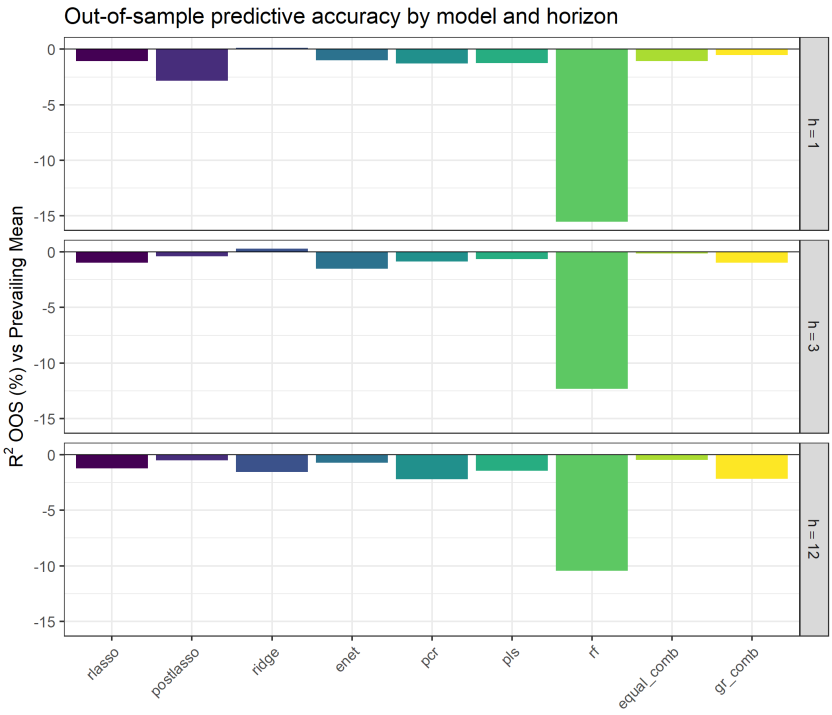


Figure 1: Out-of-sample R^2 (%) against the prevailing mean by model and horizon. Negative values indicate lower accuracy than the benchmark.

4.2 Technical indicators: Neely et al. (2014) replication and extension

To examine RQ2 directly, we adapt the Neely et al. (2014) design: expanding-window bivariate regressions of the one-month-ahead premium on one technical signal at a time, compared with the prevailing mean using a one-sided Diebold–Mariano test on squared errors. Their original design uses a 181-month initial window; with signals dated before the returns they predict, our panel provides 178 training pairs before the first forecast in 1966:01. Table 3 reports all fourteen signals.

On 1966–2011, ten of the fourteen signals have a positive R^2_{OOS}, the largest being VOL(1,12) at 0.591%, but none is significant even at 10%; the smallest p-value is 0.196 for VOL(3,12).

Table 2: Out-of-sample accuracy vs. the prevailing mean benchmark.

Model	RMSE	MAE	R^2_{OOS} (%)	HR	DM_{RMSE}	p_{DM}	p_{MAE}
Panel: $h = 1$ (OOS $n = 658$; PM hit rate 0.60)							
rlasso	4.426	3.344	-1.06	0.59	1.05	0.852	0.716
postlasso	4.465	3.348	-2.83	0.60	1.76	0.961	0.755
ridge	4.400	3.323	0.12	0.60	-0.12	0.451	0.233
enet	4.425	3.343	-1.00	0.60	1.39	0.917	0.745
pcr	4.431	3.336	-1.29	0.60	0.82	0.794	0.515
pls	4.430	3.344	-1.24	0.61	0.85	0.804	0.659
rf	4.732	3.555	-15.53	0.55	3.54	1.000	1.000
equal_comb	4.427	3.338	-1.09	0.59	0.80	0.788	0.550
gr_comb	4.414	3.326	-0.54	0.60	0.43	0.667	0.322
Panel: $h = 3$ (OOS $n = 654$; PM hit rate 0.60)							
rlasso	4.431	3.347	-0.97	0.58	1.18	0.880	0.683
postlasso	4.418	3.335	-0.38	0.60	0.46	0.677	0.415
ridge	4.403	3.329	0.30	0.60	-0.41	0.341	0.296
enet	4.442	3.366	-1.50	0.59	2.75	0.997	0.976
pcr	4.428	3.363	-0.88	0.57	0.64	0.740	0.819
pls	4.423	3.333	-0.64	0.61	0.48	0.684	0.394
rf	4.673	3.506	-12.33	0.54	2.49	0.994	0.991
equal_comb	4.412	3.338	-0.14	0.60	0.14	0.556	0.478
gr_comb	4.431	3.346	-0.97	0.59	0.87	0.809	0.624
Panel: $h = 12$ (OOS $n = 636$; PM hit rate 0.60)							
rlasso	4.470	3.391	-1.25	0.59	1.70	0.956	0.936
postlasso	4.454	3.382	-0.52	0.60	1.61	0.947	0.897
ridge	4.477	3.407	-1.54	0.59	1.98	0.976	0.984
enet	4.459	3.387	-0.72	0.59	1.29	0.902	0.893
pcr	4.491	3.404	-2.20	0.61	1.93	0.973	0.924
pls	4.474	3.394	-1.44	0.60	1.41	0.920	0.882
rf	4.669	3.560	-10.47	0.54	3.04	0.999	0.995
equal_comb	4.453	3.386	-0.48	0.58	0.89	0.814	0.853
gr_comb	4.490	3.417	-2.16	0.59	1.78	0.963	0.942

R^2_{OOS}: Campbell–Thompson (2008), in percent vs. the prevailing mean; negative values mean the model is less accurate than the benchmark. HR: hit rate, the fraction of months in which the forecast sign matches the realized sign (direction-of-change loss).

DM: Diebold–Mariano on squared errors, one-sided H1 "model more accurate"; p_{MAE}: absolute-error DM. Bartlett HAC bandwidth is $h - 1$, a maintained inference choice. $^{*}/^{**}/^{***}$: 10%/5%/1%.

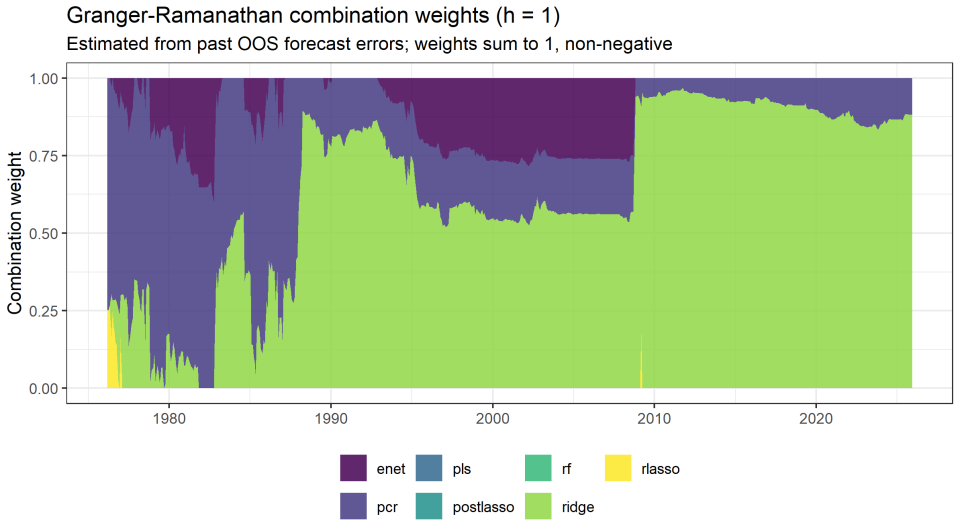


Figure 2: Constrained Granger–Ramanathan combination weights at $h = 1$, estimated from past out-of-sample forecast errors with $\sum_i w_i = 1$, $w_i \geq 0$; equal weights apply until 60 past errors are available.

Neely et al. (2014) report significant predictability for several of these indicators using the Clark–West statistic, which is designed for nested models; our Diebold–Mariano test is more conservative in this setting, so the two sets of results are not directly comparable. On the 2012–2025 extension all fourteen signals have negative R^2_{OOS}, ranging from -0.274% to -2.535%. The same picture holds in our main rolling design: over 1971–2025, all fourteen single-signal regressions estimated on the 240-month window have negative R^2_{OOS} (between -0.527% and -1.471%; smallest $p = 0.7543$), and adding the signals to the macro predictors does not help the multivariate models (Appendix A).

4.3 Economic evaluation

Table 4 reports both strategies and Figure 3 the switching wealth paths for the model with the highest R^2_{OOS} at each horizon. Because the prevailing mean is always positive, the prevailing-mean switching strategy is identical to buy-and-hold, which ends with \$341.59 per dollar at $h = 1$ net of 10 bps. The ridge strategy ends with \$499.36, with a higher Sharpe ratio (0.569 versus 0.494) and a smaller maximum drawdown (-44.4% versus -50.2%). The Granger–Ramanathan (\$421.30), PLS (\$398.41) and equal-weight (\$382.92) strategies also finish above buy-and-hold, and elastic net only marginally (\$345.52). The other four of the nine strategies finish below it: PCR (\$330.38), rlasso (\$257.78), post-LASSO (\$242.67) and the random forest (\$92.58), which switches 167 times and holds equities in only 67.8% of months. Costs matter little for slow-switching strategies: ridge falls from \$508.12 at 5 bps to \$482.26 at 20 bps.

The advantage does not survive longer horizons. At $h = 3$ only PLS (\$505.54), ridge (\$378.79) and post-LASSO (\$336.86) beat buy-and-hold (\$328.43). At $h = 12$ only PCR (\$323.26) ends above buy-and-hold (\$264.75); post-LASSO never leaves equities and matches it, and the other seven strategies end below it. Since we run nine strategies at three horizons and select the best ex post, and since we do not test whether wealth or Sharpe differences are

Table 3: Adapted Neely et al. (2014) replication and extension: expanding-window bivariate regressions on one technical signal against the prevailing mean ($h = 1$).

	1966–2011 ($n = 552$)		2012–2025 ($n = 168$)		1966–2025 ($n = 720$)	
Signal	R^2_{OOS} (%)	DM p	R^2_{OOS} (%)	DM p	R^2_{OOS} (%)	DM p
MA(1,9)	0.082	0.4439	−1.052	0.8572	−0.143	0.6117
MA(1,12)	0.433	0.2917	−1.414	0.8521	0.066	0.4617
MA(2,9)	0.166	0.4007	−0.653	0.7266	0.003	0.4977
MA(2,12)	0.569	0.2478	−1.087	0.7762	0.240	0.3708
MA(3,9)	0.254	0.3832	−1.086	0.8435	−0.013	0.5070
MA(3,12)	−0.060	0.5465	−0.776	0.9003	−0.203	0.6808
MOM(9)	−0.036	0.5324	−0.610	0.8253	−0.150	0.6548
MOM(12)	−0.058	0.5629	−0.547	0.8106	−0.155	0.6875
VOL(1,9)	0.249	0.3268	−0.859	0.7895	0.029	0.4767
VOL(1,12)	0.591	0.1998	−2.535	0.9870	−0.031	0.5201
VOL(2,9)	0.189	0.3385	−1.514	0.9524	−0.149	0.6435
VOL(2,12)	0.114	0.4025	−0.950	0.8347	−0.097	0.5917
VOL(3,9)	−0.130	0.6256	−0.274	0.6659	−0.158	0.6753
VOL(3,12)	0.531	0.1960	−0.313	0.6025	0.363	0.2553

Expanding estimation window starting with 178 correctly timed predictor–target pairs before the 1966:01 forecast. DM p: one-sided Diebold–Mariano test (squared-error loss) against the prevailing mean. Source: `output/tables/table5_neely_replication.csv`. These are our own regressions, not values from the published study.

significant, the $h = 1$ gains are suggestive at best. They also sit uneasily with the insignificant accuracy results: a switching rule depends only on the sign of the forecast, so its wealth can be driven by being out of equities in a small number of large down months.

The mean–variance results are similar in spirit. At $h = 1$ and 10 bps, six of nine models give a positive ΔU relative to the prevailing-mean investor (whose certainty-equivalent return is 7.16% p.a.), led by ridge at 1.92% p.a. and the Granger–Ramanathan combination at 1.18%; post-LASSO (−0.86%) and the random forest (−1.96%) lose. At $h = 3$ six models are again positive, and the random forest has the largest gain (2.21%) despite the worst R^2_{OOS}, plausibly because its lower average equity weight (0.761 versus 1.067 for the prevailing-mean investor) reduces portfolio variance rather than because its forecasts are more accurate. At $h = 12$ eight of nine models have negative ΔU. These utility gains are not accompanied by significant accuracy gains and are not themselves tested.

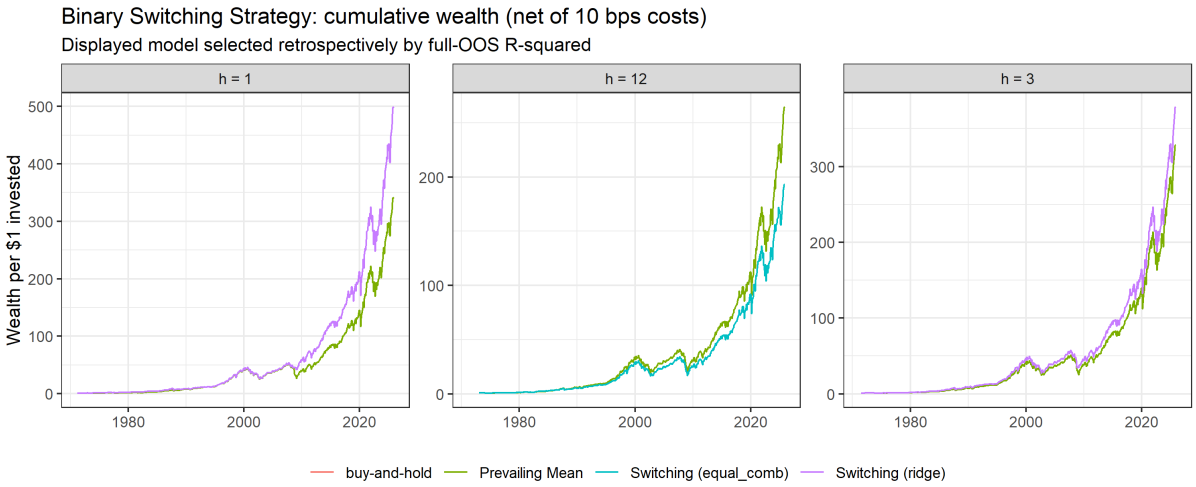


Figure 3: Cumulative wealth per \$1 of the binary switching strategy for the model with the highest R^2_{OOS} at each horizon (selected ex post), the prevailing mean and buy-and-hold, net of 10 bps per switch. Buy-and-hold coincides with the prevailing-mean strategy.

Table 4: Economic strategy outcomes net of transaction costs.

	Binary switching (10 bps)						Switching wealth		MV ΔU (% p.a.)			MV (10 bps)		
Model	Wealth	Buy-hold	Sharpe	MaxDD	Switch	Exp.	5 bps	20 bps	10 bps	5 bps	20 bps	CER	Sharpe	Mean w
Panel: $h = 1$														
rlasso	257.78	341.59	0.48	-0.44	35	0.95	262.33	248.90	0.08	0.18	-0.12	7.24	0.42	0.97
postlasso	242.67	341.59	0.46	-0.50	15	0.99	244.50	239.05	-0.86	-0.83	-0.92	6.30	0.36	0.91
ridge	499.36	341.59	0.57	-0.44	35	0.93	508.12	482.26	1.92	1.98	1.81	9.08	0.54	0.95
enet	345.52	341.59	0.51	-0.50	9	0.98	347.05	342.48	-0.23	-0.19	-0.32	6.93	0.40	0.93
pcr	330.38	341.59	0.51	-0.47	35	0.91	336.18	319.05	0.74	0.81	0.59	7.90	0.46	0.94
pls	398.41	341.59	0.54	-0.50	15	0.95	401.37	392.54	0.36	0.40	0.28	7.52	0.44	0.94
rf	92.58	341.59	0.41	-0.23	167	0.68	100.60	78.39	-1.96	-1.70	-2.47	5.21	0.27	0.79
equal_comb	382.92	341.59	0.54	-0.44	41	0.92	390.77	367.67	0.69	0.81	0.46	7.85	0.46	0.91
gr_comb	421.30	341.59	0.55	-0.44	35	0.92	428.73	406.82	1.18	1.25	1.06	8.35	0.49	0.96
Buy-and-Hold	341.59	341.59	0.49	-0.50	1	1.00								
Prevailing Mean	341.59	341.59	0.49	-0.50	1	1.00			0.00	0.00	0.00	7.16	0.42	1.07
Panel: $h = 3$														
rlasso	214.04	328.43	0.45	-0.45	25	0.96	216.71	208.78	-0.54	-0.46	-0.70	6.59	0.38	0.99
postlasso	336.86	328.43	0.50	-0.50	9	0.99	338.38	333.84	-0.35	-0.33	-0.41	6.77	0.39	0.92
ridge	378.79	328.43	0.52	-0.44	31	0.95	384.68	367.26	1.27	1.33	1.16	8.40	0.49	0.97
enet	255.73	328.43	0.47	-0.50	25	0.97	258.90	249.49	-1.80	-1.74	-1.93	5.33	0.29	0.92
pcr	220.28	328.43	0.47	-0.45	31	0.90	223.67	213.64	0.90	0.96	0.79	8.03	0.47	0.94
pls	505.54	328.43	0.57	-0.50	19	0.92	510.31	496.11	0.70	0.73	0.64	7.83	0.46	0.87
rf	212.56	328.43	0.54	-0.30	133	0.64	227.10	186.18	2.21	2.42	1.79	9.34	0.56	0.76
equal_comb	318.80	328.43	0.52	-0.44	45	0.91	326.02	304.81	0.96	1.06	0.75	8.08	0.48	0.89
gr_comb	304.91	328.43	0.51	-0.44	49	0.90	312.43	290.41	0.78	0.87	0.59	7.91	0.46	0.94
Buy-and-Hold	328.43	328.43	0.49	-0.50	1	1.00								
Prevailing Mean	328.43	328.43	0.49	-0.50	1	1.00			0.00	0.00	0.00	7.13	0.42	1.07
Panel: $h = 12$														
rlasso	148.54	264.75	0.42	-0.51	25	0.96	150.41	144.88	-0.89	-0.81	-1.05	5.72	0.34	0.93
postlasso	264.75	264.75	0.48	-0.50	1	1.00	264.89	264.49	-0.89	-0.88	-0.90	5.73	0.34	0.93
ridge	261.26	264.75	0.48	-0.50	7	0.98	262.16	259.47	-1.05	-1.02	-1.10	5.57	0.33	0.92
enet	264.29	264.75	0.49	-0.50	31	0.96	268.37	256.30	-0.42	-0.38	-0.51	6.19	0.36	0.93
pcr	323.26	264.75	0.52	-0.50	19	0.95	326.30	317.25	-0.26	-0.22	-0.35	6.35	0.37	0.93
pls	258.89	264.75	0.48	-0.50	9	0.99	260.04	256.62	-0.10	-0.07	-0.15	6.51	0.38	0.92
rf	76.28	264.75	0.38	-0.48	136	0.63	81.61	66.64	0.56	0.77	0.13	7.17	0.41	0.71
equal_comb	193.27	264.75	0.45	-0.50	39	0.95	197.05	185.92	-0.48	-0.38	-0.67	6.13	0.35	0.86
gr_comb	167.12	264.75	0.43	-0.50	45	0.91	170.91	159.80	-0.81	-0.70	-1.03	5.80	0.34	0.88
Buy-and-Hold	264.75	264.75	0.48	-0.50	1	1.00								
Prevailing Mean	264.75	264.75	0.48	-0.50	1	1.00			0.00	0.00	0.00	6.61	0.40	1.06

Switching: holds equity when the forecast is positive, monthly risk-free cash otherwise. Every strategy starts in cash, pays equity entry and subsequent turnover costs, and has no terminal liquidation. Switch counts include entry. MaxDD = maximum drawdown; Exp. = mean equity exposure; cash uses realized Rfree, not the annual tbl predictor. Each h-step forecast predicts one month's return h months ahead, not a cumulative return.
MV: mean–variance investor with $\gamma = 3$, $w_t \in [0, 1.5]$; ΔU = annualized certainty-equivalent gain over the benchmark allocation in percentage points; CER = certainty-equivalent return of the strategy (% p.a.).

4.4 Robustness

Predictor sets. Appendix Table 6 re-estimates the $h = 1$ models with technical signals only (14 columns) and macro lags only (84 columns). With technical signals only, all nine R^2_{OOS} values are negative (the best is post-LASSO at -0.53%). With macro lags only, ridge is again the only positive model (0.17%). The random forest improves markedly with the small technical set (-5.42% versus -15.53% for the full set), consistent with overfitting in the wide design.

Expanded predictors. With the 146-column expanded set, five of 27 model–horizon pairs have positive R^2_{OOS}, the largest being ridge at $h = 3$ (0.847%), with ridge at $h = 1$ only 0.029%. All models remain negative at $h = 12$ and the random forest remains the worst (-13.117% at $h = 1$). We did not run Diebold–Mariano tests for this variant, so these values are descriptive only.

Subperiods. Table 5 splits the $h = 1$ rlasso forecasts into three periods. Its R^2_{OOS} is negative in all of them and deteriorates over time, from -0.63% in 1971–1999 to -1.20% in 2000–2011 and -1.94% in 2012–2025. These comparisons are descriptive and do not identify a cause.

Switching timing. For $h > 1$, an alternative timing sets each month's position from the most recent h-month-ahead forecast (made in the previous month), so the evaluation sample shifts by $h - 1$ months. Under this timing, at $h = 3$ PLS ($387.23) and PCR ($341.14) end above the prevailing-mean strategy ($316.02), while ridge ($278.15) falls below it; at $h = 12$ only PCR ($310.82) beats the prevailing mean ($269.73). The identity of the "winning" strategy thus depends on an arbitrary timing convention, reinforcing that the economic gains are fragile.

Table 5: Descriptive LASSO out-of-sample accuracy by subperiod ($h = 1$).

Subperiod	n	First target	Last target	R^2_{OOS} (%)
1971-1999	346	1971-03-01	1999-12-01	-0.63
2000-2011	144	2000-01-01	2011-12-01	-1.20
2012-2025	168	2012-01-01	2025-12-01	-1.94

R^2_{OOS} is computed against the prevailing mean within each subperiod. These comparisons are descriptive and not causal. Source files: `forecasts_h1.csv` and `panel.csv`. Generated by `R/13_figures.R`.

5 Conclusion

RQ1. Regularized machine learning does not reliably beat the historical average. Ridge regression achieves a small positive R^2_{OOS} at $h = 1$ and $h = 3$, but no Diebold–Mariano test is significant at any horizon, and flexible nonlinear modelling (the random forest) is clearly harmful. *RQ2.* The technical signals do not add statistically significant predictive value in our adapted design: on the 1966–2011 sample most single-signal gains are positive but insignificant, after 2011 all are negative, and technical-only multivariate models are negative too. *RQ3.* Some forecasts, notably ridge, the Granger–Ramanathan combination and PLS at $h = 1$, produce higher switching wealth and positive certainty-equivalent gains net of costs, but these gains depend on model choice, horizon and timing convention, fade at $h = 12$, and are not tested for

significance. Our overall conclusion is that the prevailing mean remains a hard benchmark to beat.

The study has several limitations. We use the latest data vintage rather than real-time data, so macro predictors may contain revisions unavailable to investors at the time. Our technical signals match the published series in 99.98% of signal-months, not perfectly, and the 2012–2025 volume series is spliced from a different source. The Neely et al. (2014) analysis is an adapted, not exact, replication (different initial window, Diebold–Mariano rather than Clark–West inference). With nine forecasts, three horizons and several robustness variants, the few positive results are exactly what multiple comparisons would produce by chance, and we do not correct for this selection. The economic gains carry no significance test, and the expanded predictor set uses only eight of the Goyal et al. (2024) variables. Future work could apply formal tests of economic value, real-time vintages, and shrinkage of model forecasts towards the prevailing mean.

A Predictor-set variants

Table 6 reports R^2_{OOS} at $h = 1$ for the full 98-column design, the 14 technical signals alone and the 84 macro-lag columns alone, using the same rolling window, tuning and benchmark.

Table 6: Out-of-sample R^2 (%) at $h = 1$ by predictor set: full set (98 columns), technical signals only (14), and macro lags only (84).

Model	Full	Technical	Macro
rlasso	-1.06	-0.84	-1.03
postlasso	-2.83	-0.53	-2.80
ridge	0.12	-1.06	0.17
enet	-1.00	-1.13	-1.17
pcr	-1.29	-1.63	-2.15
pls	-1.24	-1.67	-1.25
rf	-15.53	-5.42	-15.13
equal_comb	-1.09	-1.10	-1.13
gr_comb	-0.54	-1.56	-0.49

AI use statement

Generative AI (the GitHub Copilot coding agent) was used in this project for code review and correction, for writing verification scripts that check reported numbers against the output files, and for drafting and rewriting report text, including the motivation, interpretation and discussion, at the group's request. All group members reviewed the code, results and text, and take responsibility for the content. Chat transcripts are available on request: [link/transcript to be added by group].

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